

# Regularized Linear Models

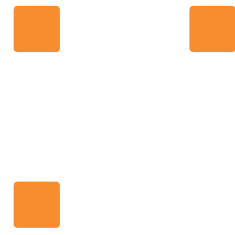


How to avoid overfitting?

# Do linear models overfit?

- Linear models are simpler than alternatives  
→ they tend to overfit less than alternatives
- They even often underfit when:
  - `n_features` is small (e.g. less than 10 features)
  - the problem is not linearly separable

But...



# Linear models can also overfit!

Possible causes:

- too few samples  $n_{\text{samples}} \ll n_{\text{features}}$
- Many uninformative features

# Linear models can also overfit!

Possible causes:

- too few samples  $n_{\text{samples}} \ll n_{\text{features}}$
- Many uninformative features

Example for linear regression:

```
Sale_Price = 0.1 * Gr_Liv_Area  
            + 1.1 * Year_Built  
            - 8.9 * Full_Bath  
            + 2.5 * First_Owner_Is_Born_In_January  
            - 1.5 * First_Owner_Is_Born_In_February  
            ...  
            - 2200.0
```

# Regularization can reduce overfitting

Unregularized regression:

```
from sklearn.linear_model import LinearRegression

linear_regression = LinearRegression()
linear_regression.fit(X, y)
```

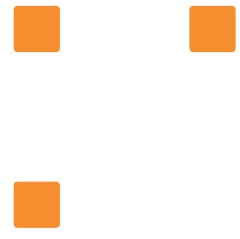
Ridge regression:

```
from sklearn.linear_model import Ridge

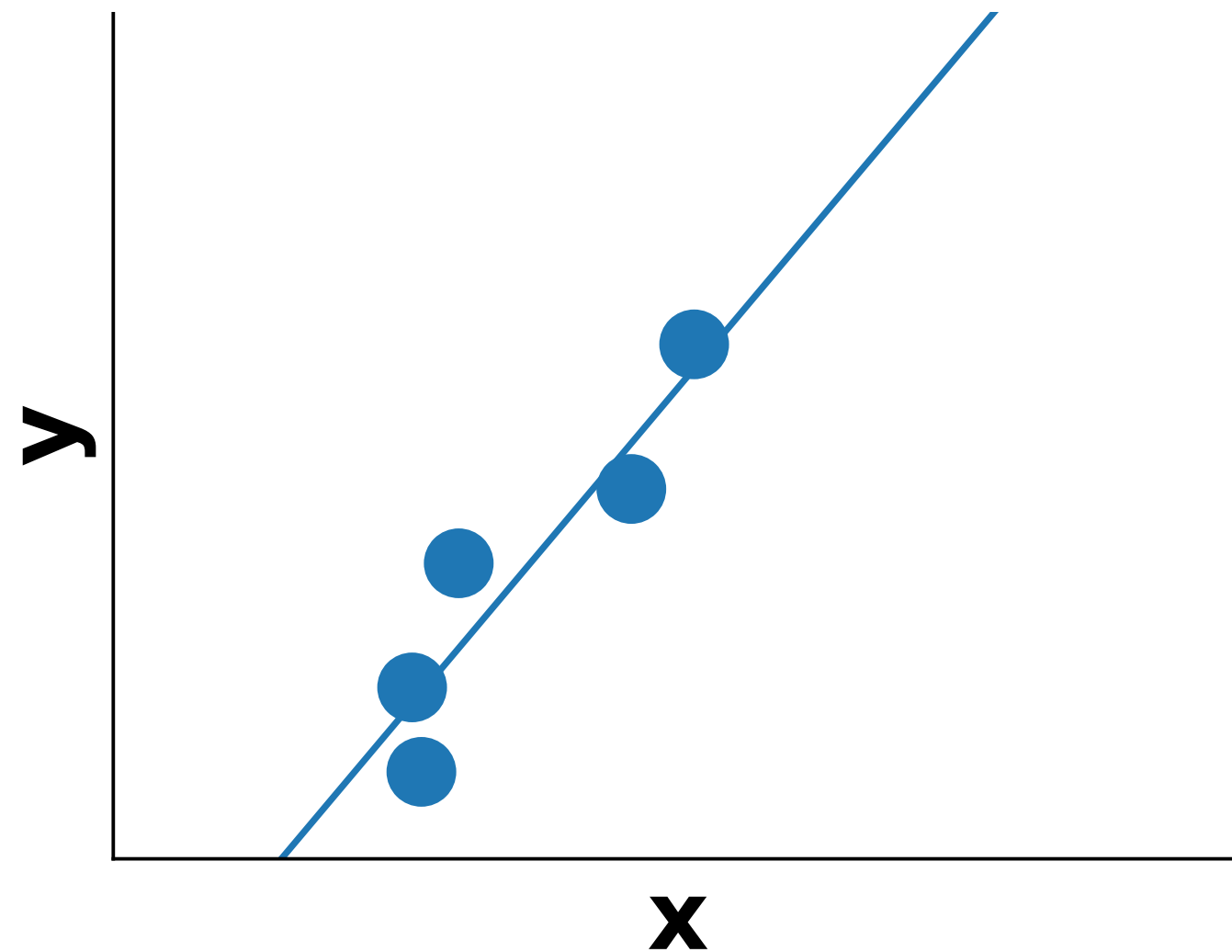
ridge_regression = Ridge(alpha=0.01)
ridge_regression.fit(X, y)
```

- Ridge regression pulls the coefficients towards 0.
- Large alpha  $\rightarrow$  more regularization
- Always use Ridge with a carefully tuned alpha!





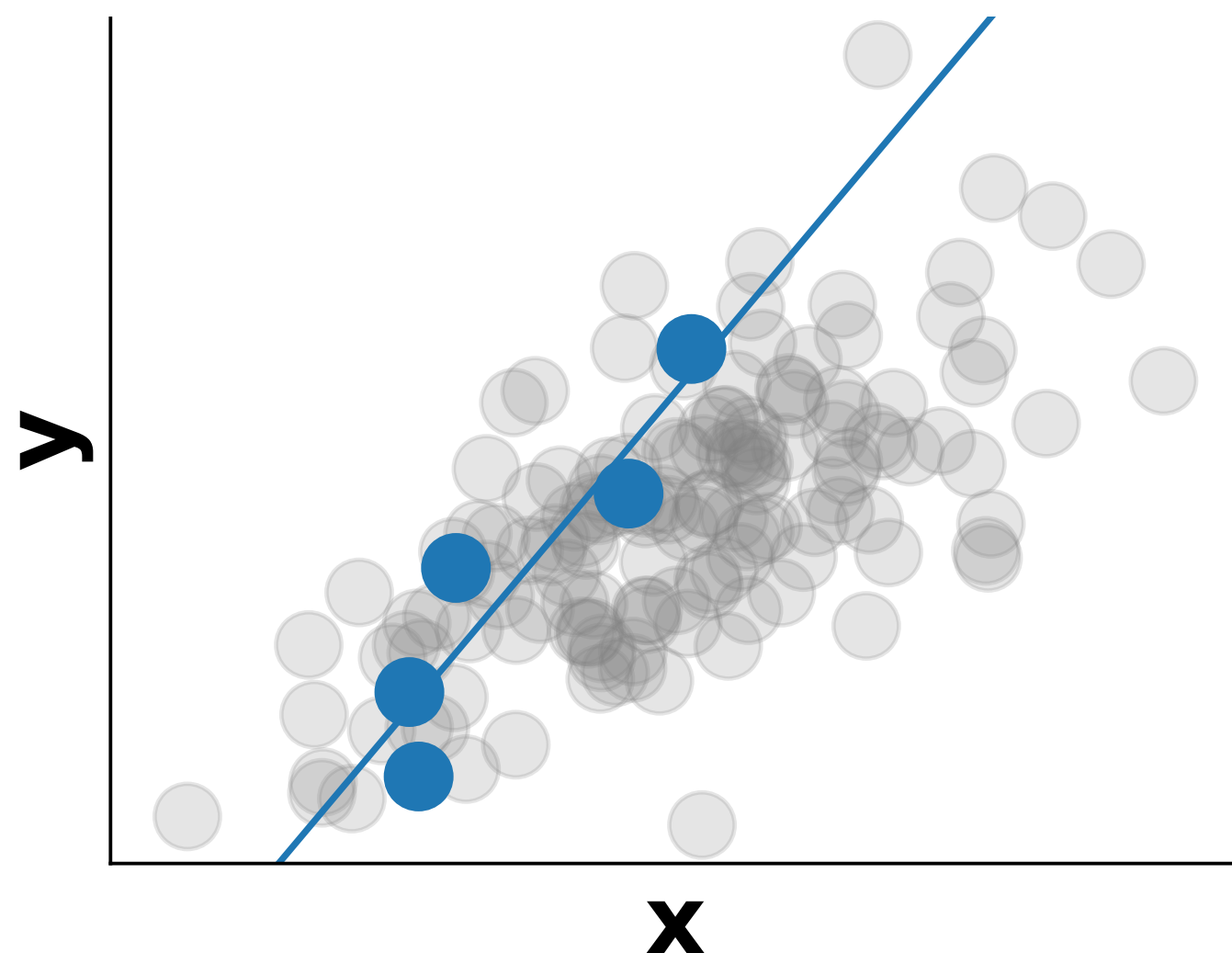
# Regularization on a simple example



Small training set

Fit a linear model without regularization

# Regularization on a simple example

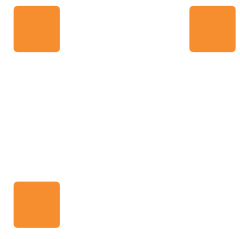


Small training set

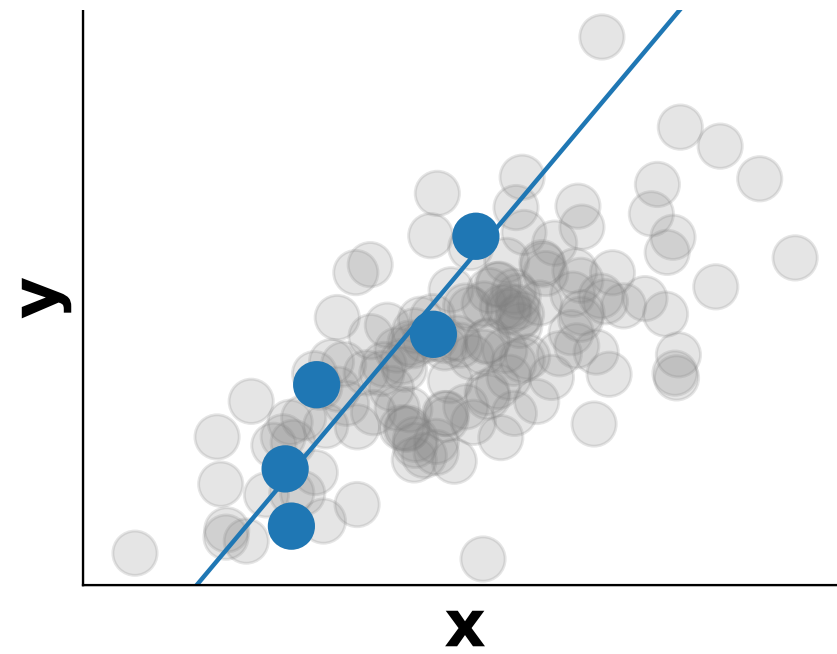
Fit a linear model without regularization

Training points sampled at random

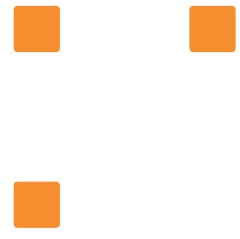
Can overfit if the data is noisy!



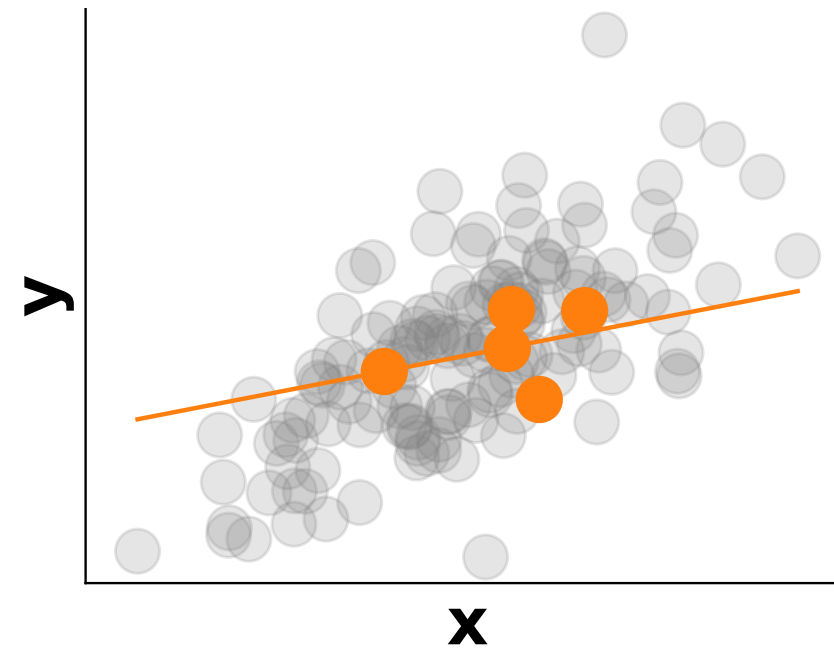
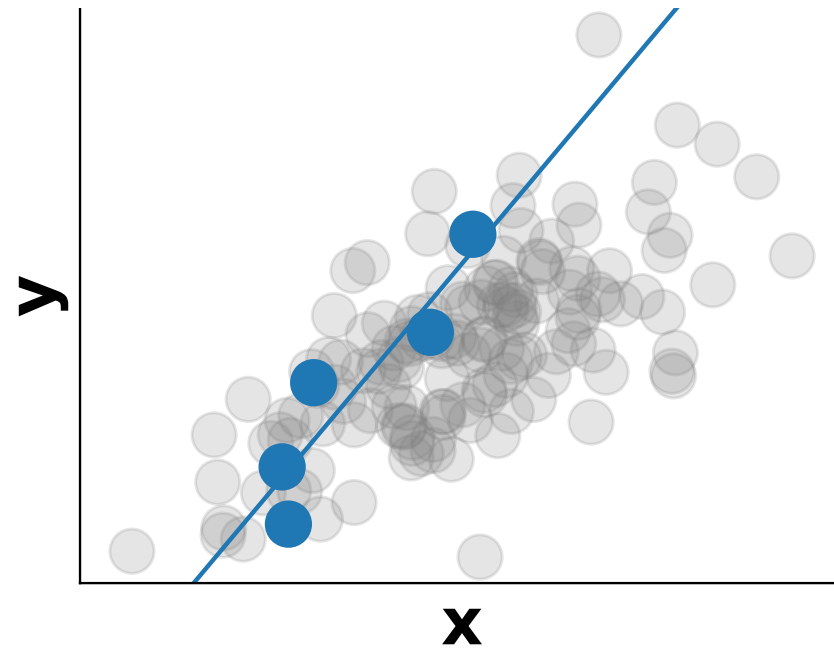
# Regularization on a simple example





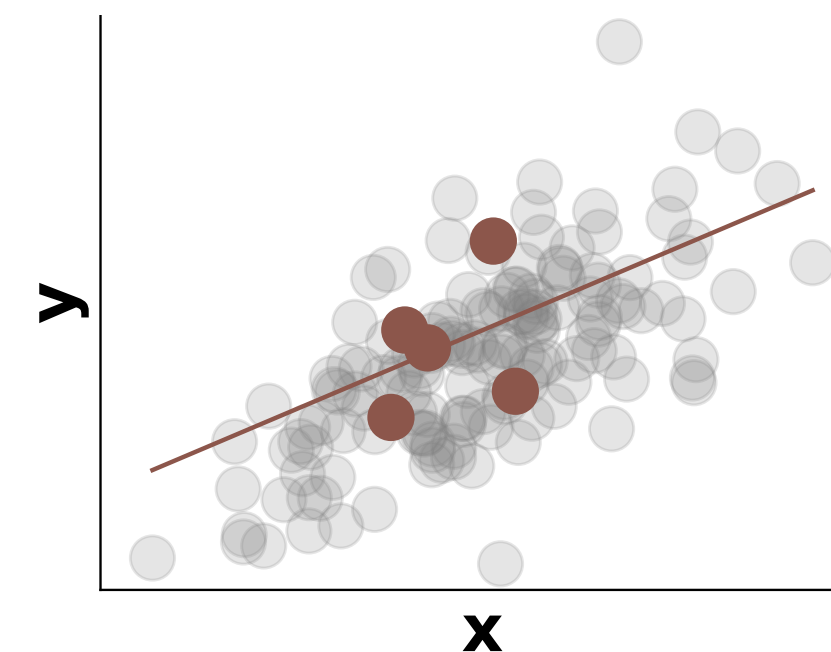
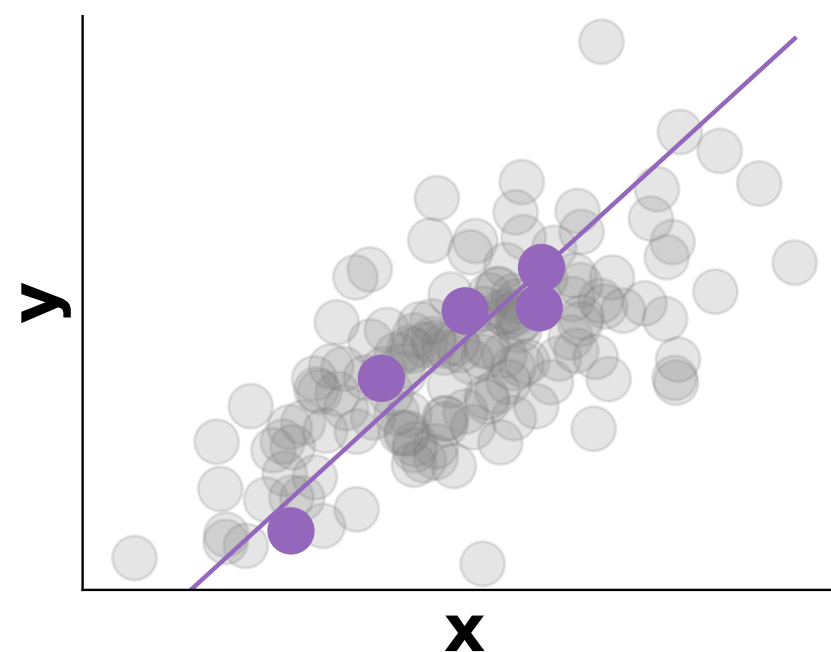
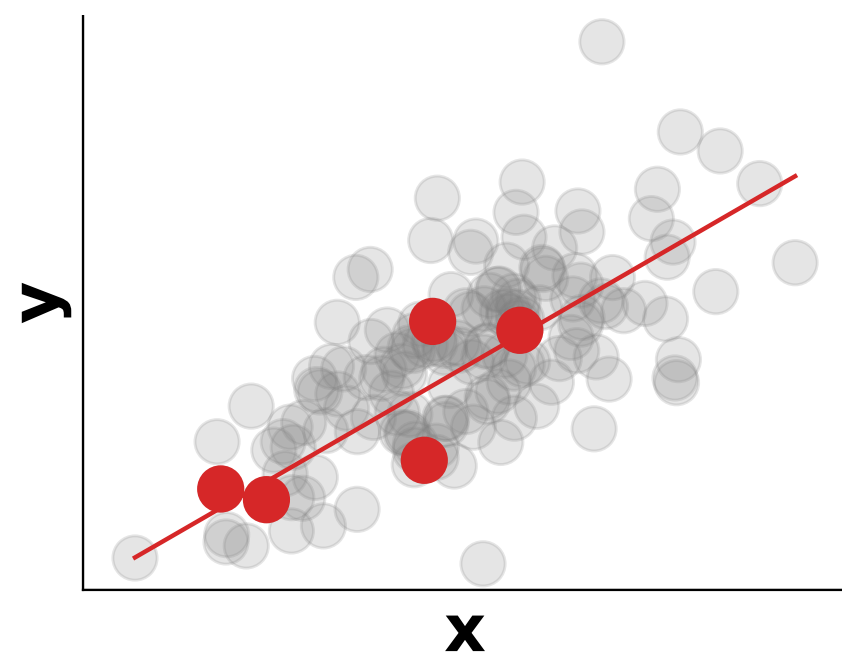
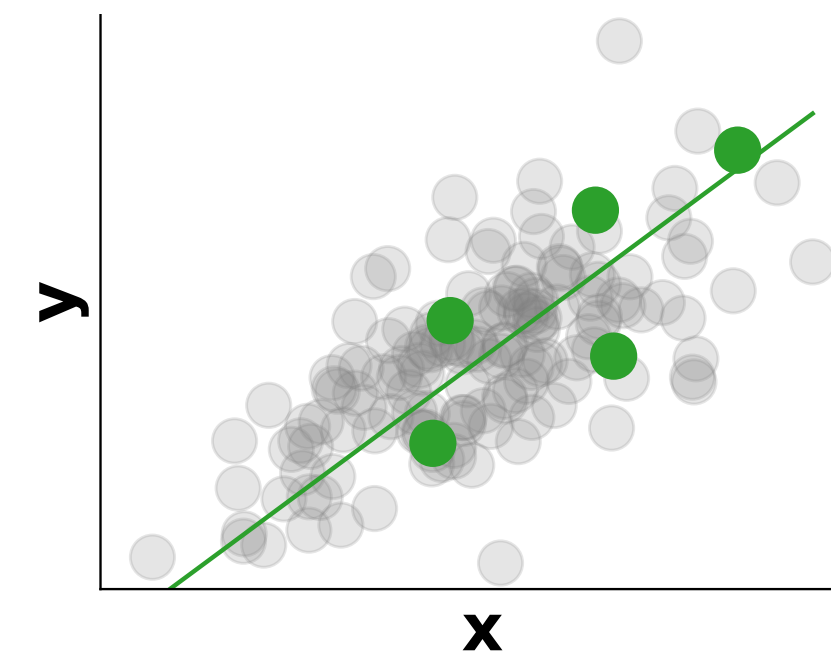
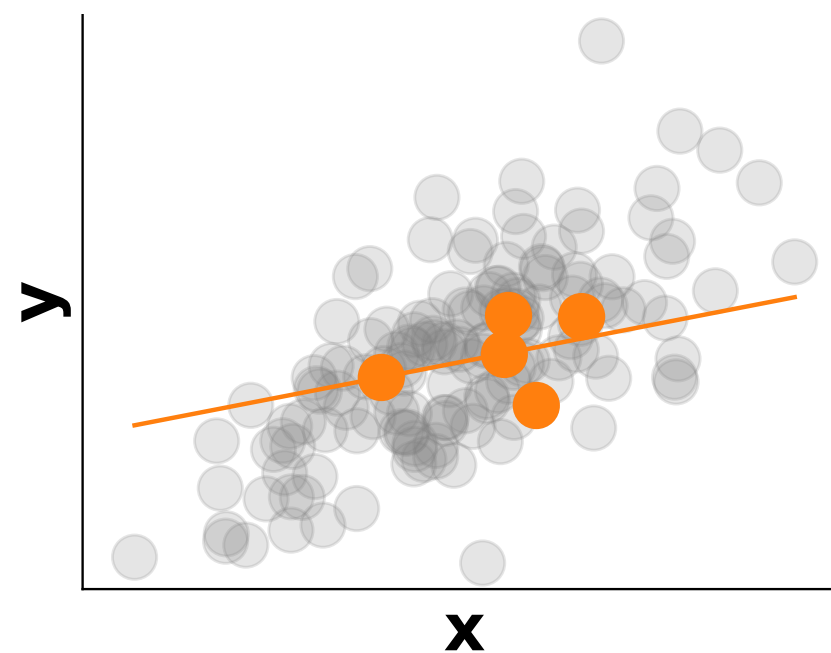
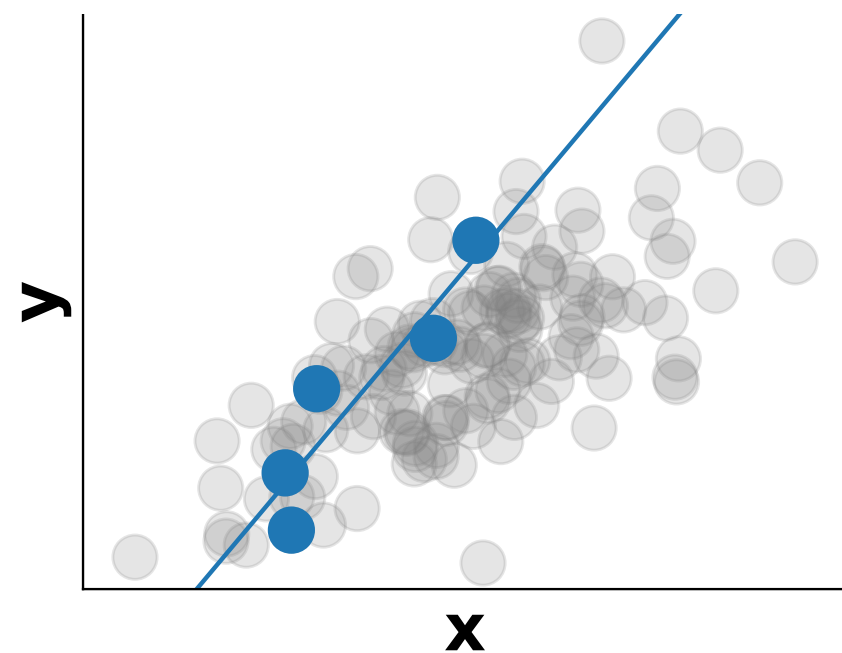


# Regularization on a simple example

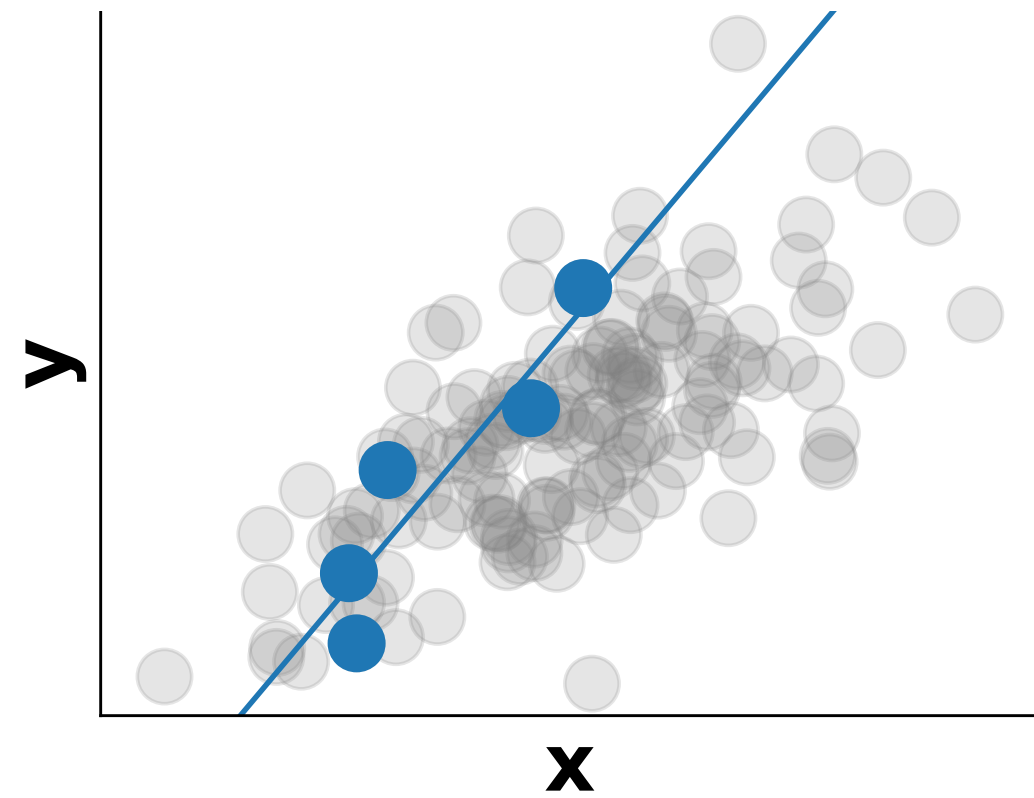




# Regularization on a simple example



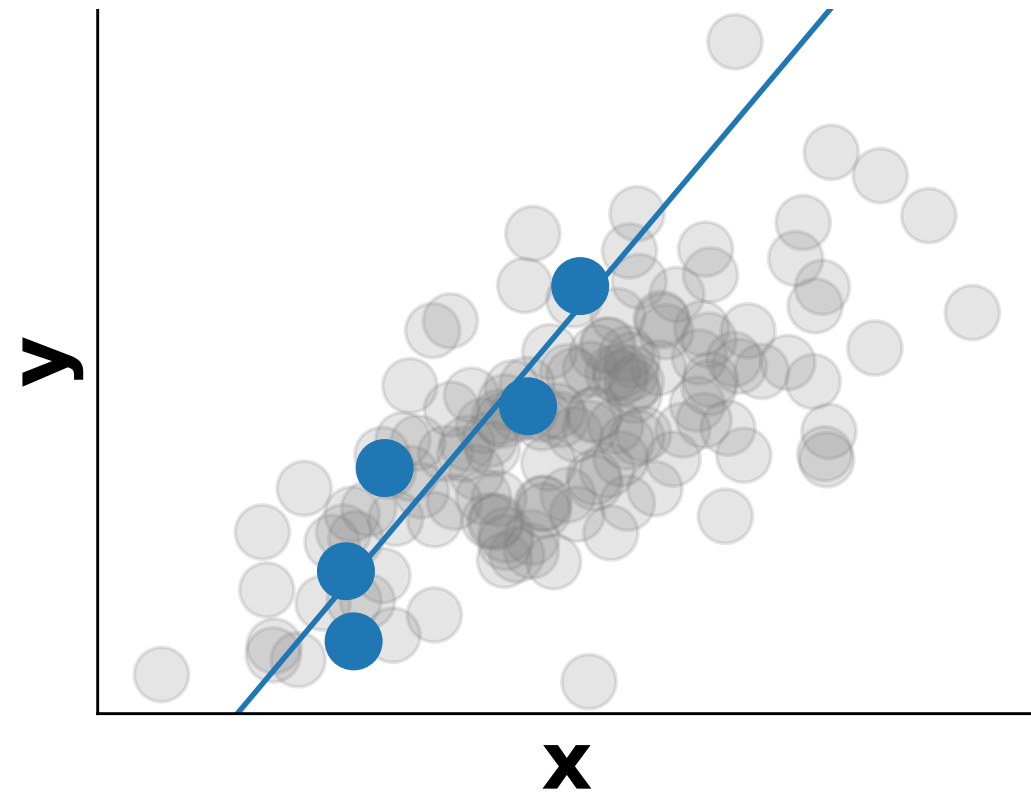
# Bias-variance tradeoff in Ridge regression



LinearRegression  
(no regularization)

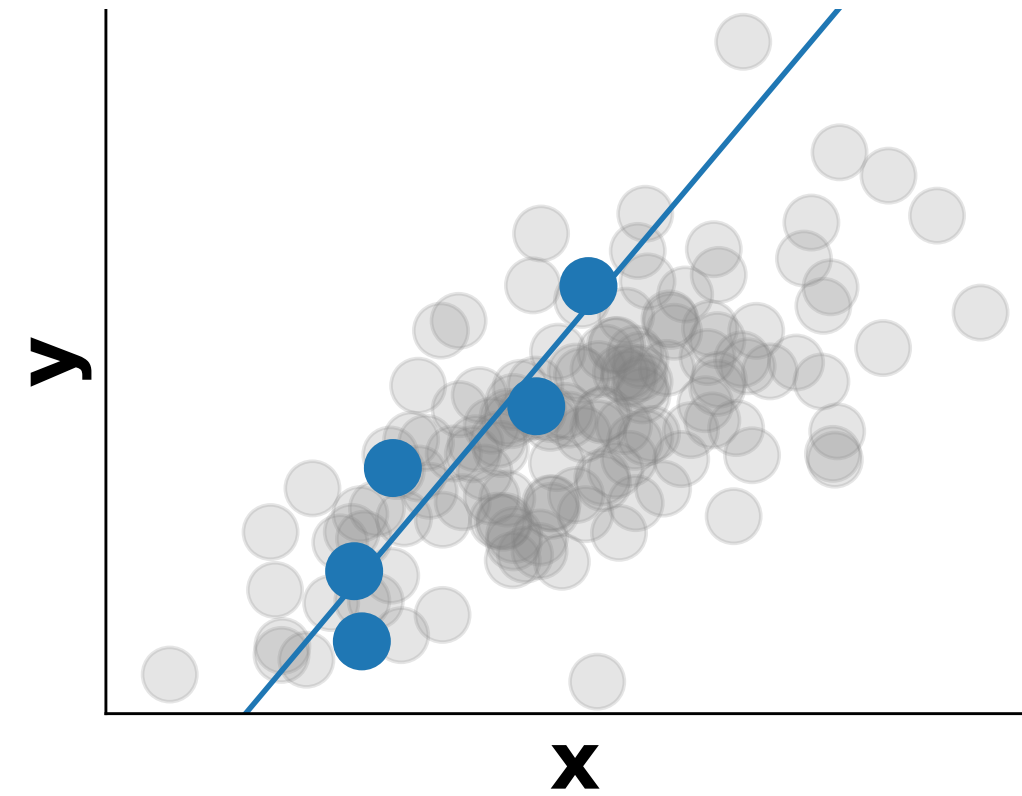
High variance, no bias

# Bias-variance tradeoff in Ridge regression



LinearRegression  
(no regularization)

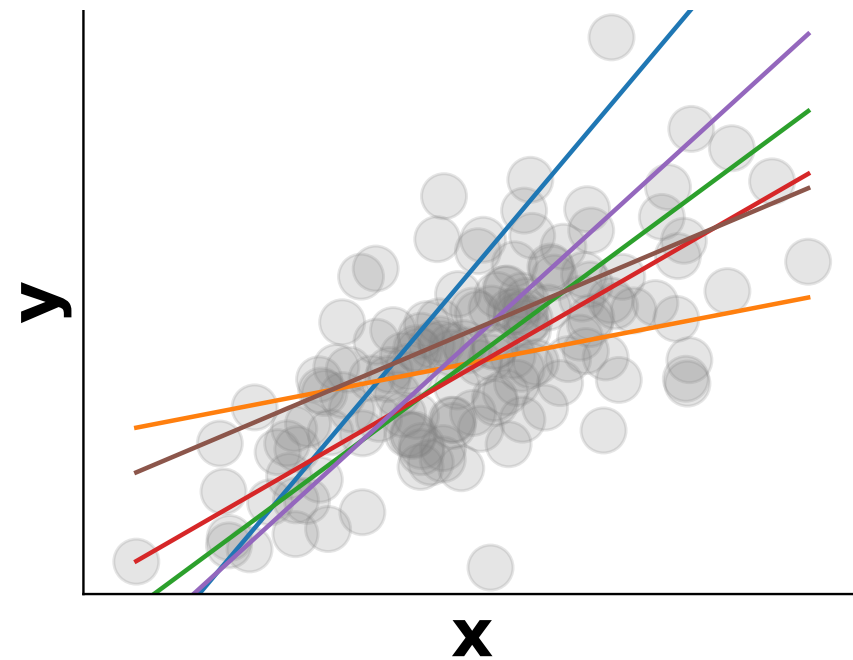
High variance, no bias



Ridge regression  
(regularized)

Lower variance, but biased

# Bias-variance tradeoff in Ridge regression

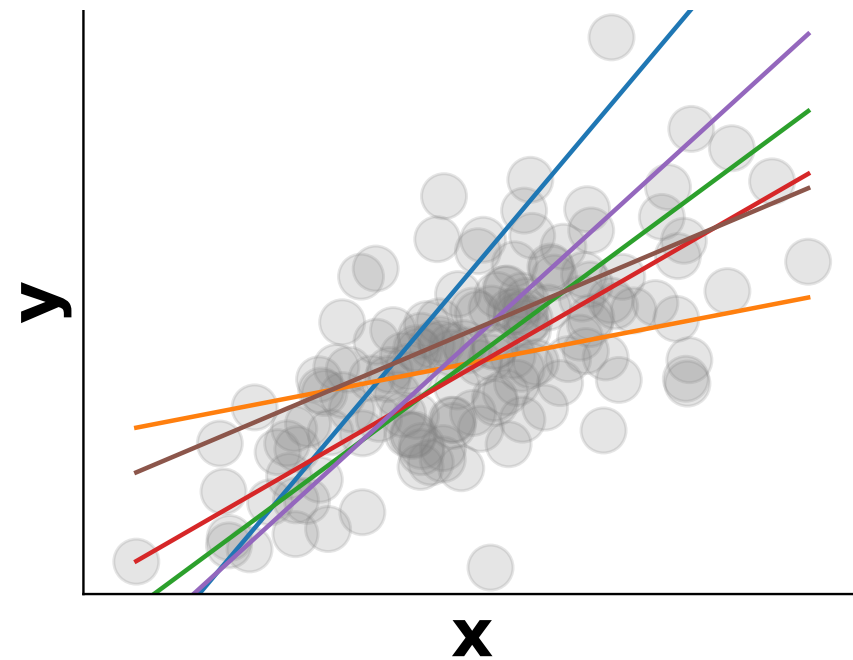


Too much variance

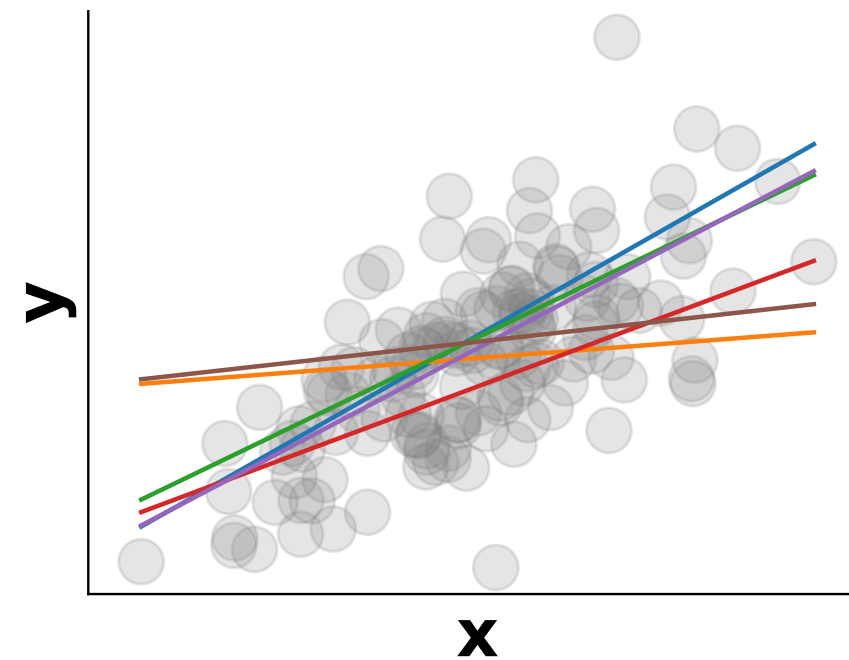
alpha too small



# Bias-variance tradeoff in Ridge regression

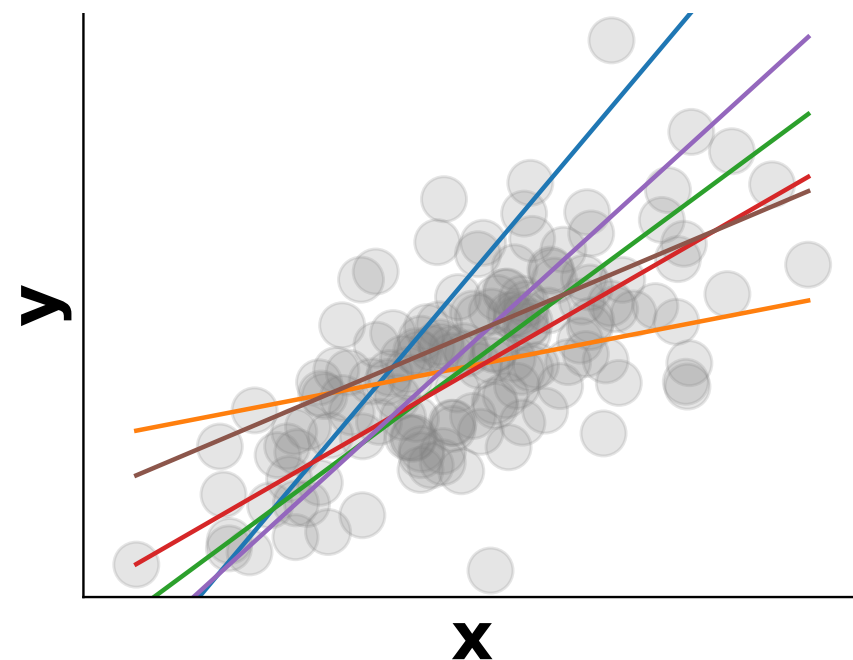


Too much variance  
alpha too small

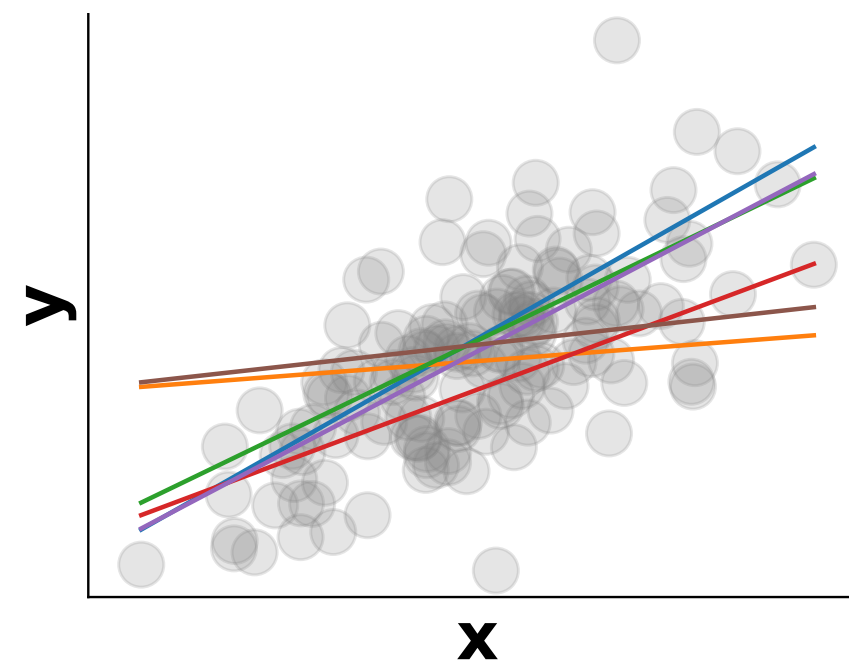


Best tradeoff

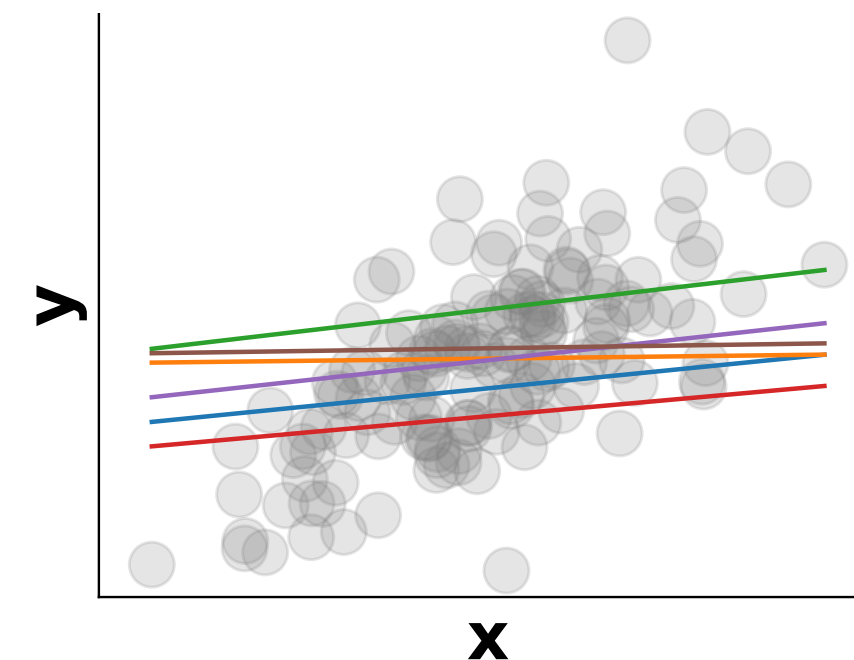
# Bias-variance tradeoff in Ridge regression



Too much variance  
alpha too small



Best tradeoff



Too much bias  
alpha too large

# Automated tuning for regularization

```
from sklearn.linear_model import Ridge
from sklearn.model_selection import GridSearchCV

param_grid = {
    "alpha": [0.001, 0.1, 1, 10, 1000],
}
model = GridSearchCV(Ridge(), param_grid)
model.fit(X, y)

print(model.best_parameters_)
```



# Automated tuning for regularization

```
from sklearn.linear_model import Ridge
from sklearn.model_selection import GridSearchCV

param_grid = {
    "alpha": [0.001, 0.1, 1, 10, 1000],
}
model = GridSearchCV(Ridge(), param_grid)
model.fit(X, y)

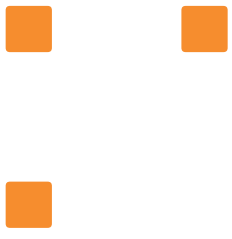
print(model.best_parameters_)
```

```
from sklearn.linear_model import RidgeCV

model = RidgeCV(
    alphas=[0.001, 0.1, 1, 10, 1000]
)
model.fit(X, y)

print(model.alpha_)
```

Almost as fast as fitting a  
single Ridge model



# Regularization in logistic regression

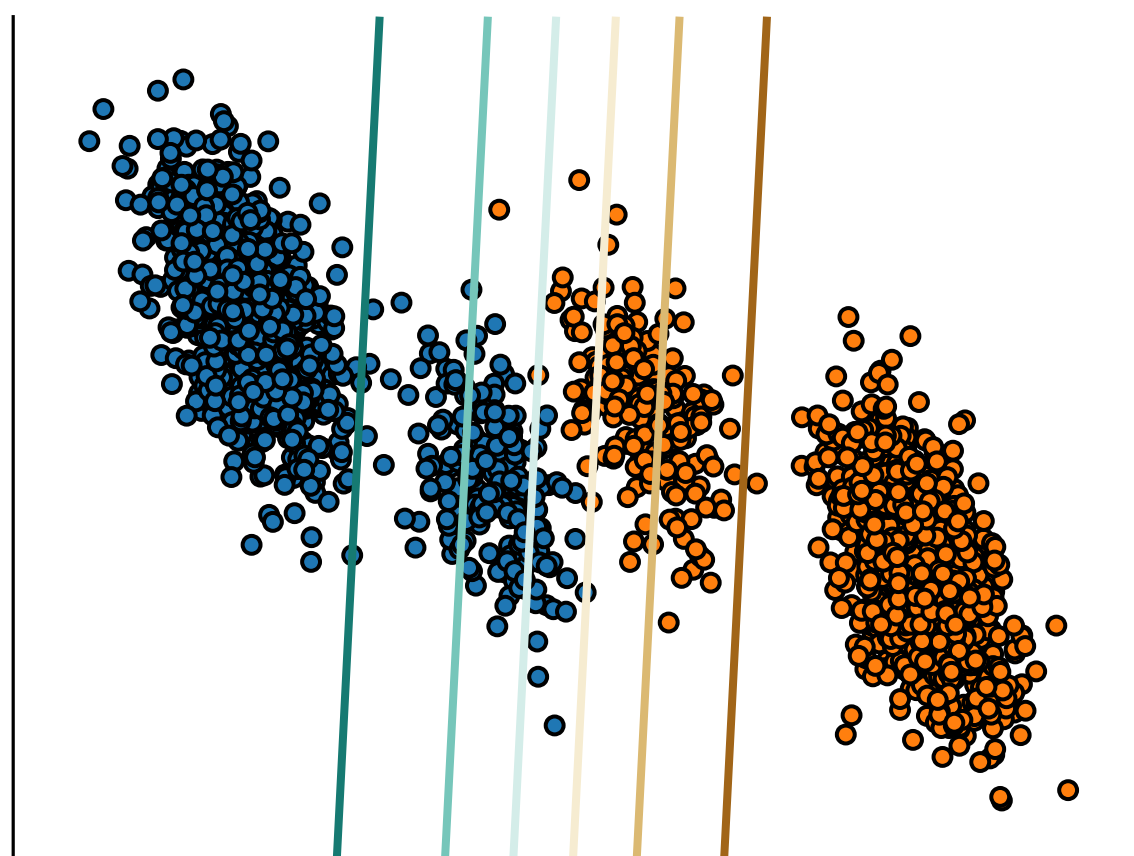


`LogisticRegression(C=1)` is regularized by default!

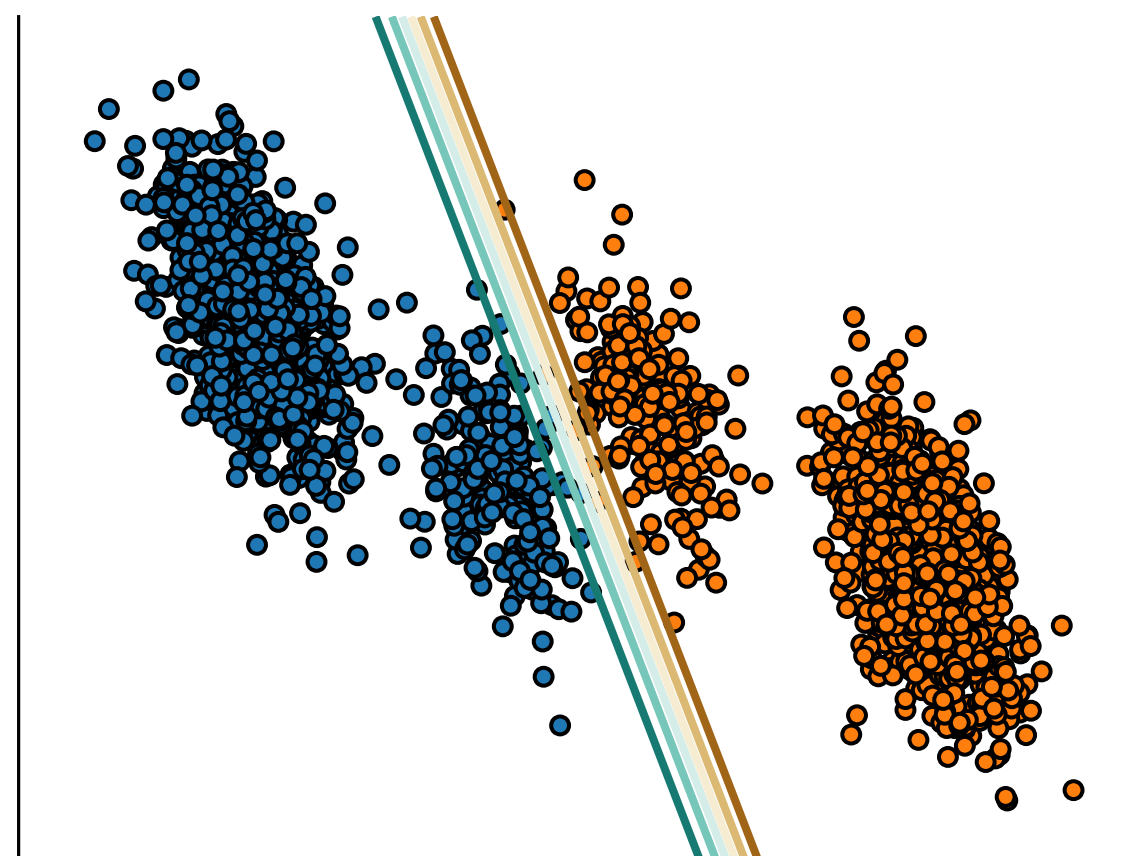
# Regularization in logistic regression

`LogisticRegression(C=1)` is regularized by default!

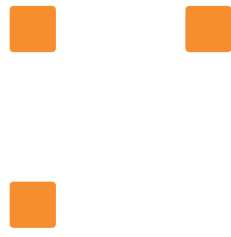
High C value  $\rightarrow$  weaker regularization.



Small C



Large C



# Take home messages on linear models

- Can overfit when:
  - `n_samples` is too small and `n_features` is large
  - presence of non-informative features

# Take home messages on linear models

- Can overfit when:
  - `n_samples` is too small and `n_features` is large
  - presence of non-informative features
- Regularization for regression:
  - linear regression → ridge regression
  - large alpha parameter → strong regularization



# Take home messages on linear models

- Can overfit when:
  - `n_samples` is too small and `n_features` is large
  - presence of non-informative features
- Regularization for regression:
  - linear regression → ridge regression
  - large alpha parameter → strong regularization
- Regularization for classification:
  - logistic regression regularized by default
  - small C parameter → strong regularization